

SATQUERY AI — REMOTE SENSING MISSION REPORT

Indian Space Research Organisation (ISRO) | SIH 26167 Intelligence Dispatch

Natural Language Query:	How healthy is the vegetation and crops?
Identified Workflow Intent:	VQA
Dispatched Specialist:	VQAEngine
Composite Confidence:	94% (HIGH)

1. Executive Analysis & Findings

Assessment: Vegetation cover spans 45.4% (1190.1 hectares / ~11.9 km²) of the surveyed area, predominantly concentrated in the North-East sector. Mean estimated NDVI proxy is +0.23, indicating moderate photosynthetic activity / mixed crop.

- Dominant Terrain Category: Vegetated & Forested Landscape (45.4% coverage)
- Total Survey Extent: 26.21 km² (2621.4 hectares) at 10.0m GSD
- Spatial Indices: NDVI=+0.23 (Vegetation vigor) | NDWI=+0.34 (Water absorption)
- Quadrant Distribution: Built-up peaked in North-West, Vegetation peaked in North-East
- Spatial Reference System: EPSG:32643 (ISRO WGS 84 / UTM)

2. 4-Signal Multi-Criteria Confidence Matrix

Metric Signal	Component	Score (%)	Status
C_model	Specialist Model Inference Certainty	93%	Optimal
C_sensor	Sensor Radiometric SNR & Quality	90%	Optimal
C_alignment	Spatial Co-Registration (CRS & IoU)	98%	Verified
C_resolution	Ground Sample Distance Suitability	95%	Optimal

3. Auditable Agent Execution Pipeline Trace

Step	Action / Milestone	Tool Executed	Latency	Status
1	Raster Ingestion & Metadata Parsing	GeoTIFFProcessor	1.7 ms	COMPLETED
2	Agentic Query Classification & Tool Rout	AgentClassifier	0 ms	COMPLETED
3	Specialist Model Execution (VQAEngine)	VQAEngine	145.1 ms	COMPLETED
4	Evidence Fusion & 4-Signal Confidence Ra	CompositeConfidenceEngine	0 ms	COMPLETED
5	Plain-English Simplification & Insight S	Smart Explainer	0 ms	COMPLETED